

Meat-Related Compounds and Colorectal Cancer Risk by Anatomical Subsite

Since meat may be involved in the etiology of colorectal cancer, associations between meat-related compounds were examined to elucidate underlying mechanisms in a population-based case-control study. Participants (989 cases/1,033 healthy controls) completed a food frequency questionnaire with a meat-specific module. Multivariable logistic regression was used to examine associations between meat variables and colorectal cancer; polytomous logistic regression was used for subsite-specific analyses. The following significant positive associations were observed for meat-related compounds: 2-amino-3,4,8-trimethylimidazo[4,5-f]quinoxaline (DiMeIQx) and colorectal, distal colon, and rectal tumors; 2-amino-3,8-dimethylimidazo[4,5-f]quinoxaline (MeIQx) and colorectal and colon cancer tumors; nitrites/nitrates and proximal colon cancer; 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) and rectal cancer; and benzo[a]pyrene and rectal cancer (P-trends < 0.05). For analyses by meat type, cooking method, and doneness preference, positive associations between red processed meat and proximal colon cancer and pan-fried red meat and colorectal cancer were found (P-trends < 0.05). Inverse associations were observed between unprocessed poultry and colorectal, colon, proximal colon, and rectal tumors; grilled/barbequed poultry and proximal colon cancer; and well-done/charred poultry and colorectal, colon, and proximal colon tumors (P-trends < 0.05). HCAs, PAHs, nitrites, and nitrates may be involved in colorectal cancer etiology. Further examination into the unexpected inverse associations between poultry and colorectal cancer is warranted.